

Ahnaf An Nafee

Computer Science PhD Student | George Mason University

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Computer Science PhD student focused on human-computer interaction, with industry experience in platform engineering and mobile/backend product delivery. At the DCXR Lab, I formulate hypotheses, design and pilot-test experiments, and integrate Unity/C# and Python study systems. Former CTO seeking Summer 2027 research internships in HCI, human-centered AI, graphics, and ML systems.

[01] EDUCATION

George Mason University

AUG 2025 - PRESENT

PhD in Computer Science, in progress

Fairfax, VA

Anticipated graduation: June 2029. Research focus: human-computer interaction and interactive systems. DCXR Lab researcher.

Drexel University

SEP 2018 - JUN 2022

BS in Computer Science, Magna Cum Laude

Philadelphia, PA

Concentrations in Artificial Intelligence and Game Programming & Development. Four-year co-op program.

[02] RESEARCH EXPERIENCE

DCXR Lab, George Mason University

JAN 2026 - PRESENT

Researcher; Graduate Research Assistant appointment in Summer 2026

Fairfax, VA

- Formulate hypotheses and design experiments for IRB-approved human-subjects research; integrate and pilot-test Unity/C# and Python study systems for fNIRS-informed classification, synchronized telemetry, preprocessing, and data-quality checks.

[03] SELECTED PROJECTS AND EVALUATION

Player 2

ONGOING

Continuing independent development

Mobile, web, and backend

- Evolved a nine-repository iOS, Android, and web platform using Java/Spring, PostgreSQL/Redis, cloud infrastructure, release automation, and observability.
- Built feed, matchmaking, and discovery ranking with attributed feedback, Python/LightGBM training, and Java shadow scoring; learned-model ranking remains shadow-only.
- Benchmarked one million synthetic users; removed full-table and N+1 work to cut warm-feed time from 7.4 seconds to 0.70-1.69 seconds and raise modeled capacity from 39 to 330 requests/second.
- Added 15-locale and right-to-left support; reduced feed p99 frame time from 101 ms to 36 ms and jank from 9.6% to 4.1%.

Offline recommendation studies

2026

Independent research projects

Python, PyTorch, LightGBM, MovieLens 1M

- Evaluated temporal LambdaRank against validation-selected popularity on 961 held-out users (NDCG@10 0.108 vs. 0.190; paired difference -0.082, 95% interval [-0.093, -0.071]); kept the higher-scoring test-only item-kNN result descriptive.
- Compared PyTorch GRU and causal Transformer next-rating rankers (Transformer-minus-GRU +0.014 NDCG@10; 95% paired user-bootstrap interval [+0.008, +0.020]); built temporal-leakage, cold-history, artifact, and CPU/GPU cost checks. Outcomes reflect rating activity, not engagement.

[04] INDUSTRY EXPERIENCE

Mindex

FEB 2023 - AUG 2025

DevOps Engineer, contractor for Paychex under Mindex

United States

- Standardized Kubernetes/OpenShift deployment patterns across 10+ service teams, establishing a consistent path for production releases.
- Designed and rolled out OpenShift role-based access controls with security and infrastructure partners to address privilege-escalation risk on a shared platform.
- Automated the TLS certificate lifecycle, eliminating manual renewals and reported certificate-expiration downtime.
- Built Gradle tooling for container-image certification, reducing CI/CD congestion and contributing to a reported 10% reduction in annual operating expense.

Dynasty 11 Studios

JUN 2022 - FEB 2023

Chief Technology Officer

United States

- Promoted to CTO within nine months; owned Player 2's technical roadmap and led a distributed team across React Native, Java/Spring, PostgreSQL, AWS, and product delivery.
- Advanced the award-winning prototype toward production readiness across matchmaking, chat, community, and marketplace features; reduced cloud costs by 80% through architecture and delivery changes.

Dynasty 11 Studios

SEP 2021 - JUN 2022

Software Engineer

United States

- Built the React Native and Java/Spring foundation across authentication, STOMP/WebSocket chat, matchmaking, PostgreSQL services, and repeatable mobile/backend releases.

PHL Collective

MAR 2021 - SEP 2021

Technical Engineer

United States

Contributed Unity/C# engineering and real-time technical work within a game-development team.

giffot.ai

FEB 2020 - MAR 2020

UX Designer

Early-stage product team

Contributed user-experience design work to an early-stage product team.

[05] TEACHING EXPERIENCE

George Mason University

AUG 2026 - PRESENT

Graduate Teaching Assistant

Fairfax, VA

PRIOR APPOINTMENT: AUG 2025 - MAY 2026

Support computer science coursework and labs through instruction, debugging, and student mentorship.

Drexel University

SEP 2021 - JUN 2022

Teaching Assistant

Philadelphia, PA

Supported programming assignments, course concepts, and technical problem-solving.

[06] TECHNICAL SKILLS AND RECOGNITION

Programming: Python, Java, Go, C++, C#, Rust, TypeScript, JavaScript, SQL, Bash.

Research and ML tools: Unity, NumPy, pandas, SciPy, scikit-learn, LightGBM; research instrumentation, data-quality validation, offline evaluation, synthetic load testing.

Platforms: React Native, React, Java/Spring, PostgreSQL, Redis, AWS, Kubernetes, OpenShift, Docker, Terraform, GitHub Actions, Jenkins, Gradle, Cloudflare.

Recognition: Drexel Senior Project Showcase Winner for Player 2; Dean's List; Founder's Scholarship.

Languages: English (full professional proficiency); Bangla (native).